AP CALCULUS AB

Unit 1 • Lesson 1.8 practice • Answer key

Lesson 1.8 • Determining Limits Using the Squeeze Theorem

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to 0} x \sin \frac{1}{x}$ is

(A) 0

(B) 1

(C) It does not exist.

(D) ∞

Solution

Answer: (A). $-|x| \leq x \sin \frac{1}{x} \leq |x|$, and both bounds approach 0.

Question 2

NO CALCULATOR

$\lim_{x \to 0} \frac{\sin 4x}{2x}$ is

(A) 4

(B) 2

(C) $\frac{1}{2}$

(D) 1

Solution

Answer: (B). $\frac{\sin 4x}{2x} = 2 \cdot \frac{\sin 4x}{4x} \to 2.$

Question 3

NO CALCULATOR

$\lim_{x \to 0} \frac{1 - \cos x}{x}$ is

(A) 1

(B) -1

(C) 0

(D) It does not exist.

Solution

Answer: (C). This is one of the two standard trigonometric limits.

Question 4

NO CALCULATOR

If $3 - x^2 \le f(x) \le 3 + x^2$ for all x, then $\lim_{x \to 0} f(x)$ is

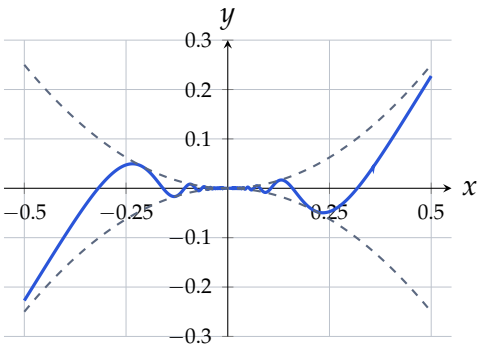
(A) 0
(B) 6
(C) It cannot be determined.
(D) 3

Solution
Answer: (D). Both bounds approach 3; by the squeeze theorem so does f.

Question 5 (graph)

NO CALCULATOR

The graph of $f(x) = x^2 \sin \frac{1}{x}$ is shown between the dashed graphs of $y = x^2$ and $y = -x^2$. What is $\lim_{x \to 0} f(x)$?



(A) 0
(B) 1
(C) It does not exist because f oscillates.
(D) -1

Solution
Answer: (A). f oscillates, but it is trapped between $-x^2$ and x^2, which both approach 0. By the squeeze theorem the limit is 0.

Question 6 (Free response)

NO CALCULATOR

(a) Explain why $-|x| \le x \cos \frac{1}{x} \le |x|$ for $x \neq 0$, and use it to find $\lim_{x \to 0} x \cos \frac{1}{x}$.
(b) Find $\lim_{x \to 0} \frac{\tan x}{x}$.
(c) Find $\lim_{x \to 0} \frac{x}{\sin 3x}$.

Solution
(a) $-1 \le \cos \frac{1}{x} \le 1$; multiplying by $|x| \ge 0$ gives $-|x| \le |x| \cos \frac{1}{x} \le |x|$, and $x \cos \frac{1}{x}$ equals $\pm|x| \cos \frac{1}{x}$, so it lies between $-|x|$ and $|x|$. Both bounds approach 0, so by the squeeze theorem the limit is 0.
(b) $\dfrac{\tan x}{x} = \dfrac{\sin x}{x} \cdot \dfrac{1}{\cos x} \to 1 \cdot 1 = 1.$
(c) $\dfrac{x}{\sin 3x} = \dfrac{1}{3} \cdot \dfrac{3x}{\sin 3x} \to \dfrac{1}{3}.$